VP's Kamalnayan Bajaj Institute of Engineering and Technology, Baramati

Department of Artificial Intelligence and Data Science Engineering

Academic Year: 2024-25

Subject: Database Management System

Class: S.YBTech.

QUESTION BANK LIST

UNIT NO. I

1. Define DBMS. Explain advantages of DBMS over file system
2. Elaborate the need of database views. Also explain the situations where in the view created are updateable views.
3. Explain data abstraction in detail.
4. Explain data independence in detail.
5. What is database architecture and its application?
6. What is data model and explain it in detail?
7. What is key. Explain all types of keys.
8. Explain difference between: File system and DBMS, RDBMS and DBMS
9. A weak entity set can always be made into strong entity set by adding to its attributes; the primary key attributes of its identifying entity set. Outline what sort of redundancy will result if we do so while converting into tables.
10. Draw ER diagram for following system: College Management System and Hospital Management System

UNIT NO. II

1. Explain the difference between DDL and DML with examples.
2. Describe the use of SQL clauses in a SELECT statement.
3. Discuss the use of indexes and sequences in SQL.
4. Explain the concept and use of views in SQL.
5. Describe different types of joins used in SQL queries.
6. Illustrate the concept of nested queries with a simple example.
7. Differentiate between stored procedures and functions in PL/SQL.
8. Explain the working of cursors in PL/SQL.
9. Describe how triggers are used to maintain data integrity.
10. Discuss the purpose of roles and privileges in SQL.

UNIT NO.III

1. Explain the concept of the relational model with an example.
2. Describe Codd's Rules and their significance.
3. Discuss the role of domain and referential integrity in relational databases.
4. Explain the features of a good relational database design.
5. Describe the process of normalization and its advantages.
6. Illustrate First Normal Form (1NF) with an example.
7. Explain the concept of functional dependency.
8. Describe how to decompose a relation using functional dependencies.
9. Compare 2NF, 3NF, and BCNF with suitable examples.
10. Explain the need for Fourth Normal Form (4NF).
11. **Solve examples on Functional Dependency**

UNIT NO.IV

1. Define a database transaction and explain its properties.
2. Describe the different states of a transaction.
3. Explain the concept of a serial schedule in transaction management.
4. Differentiate between conflict and view serializability.
5. Describe what is meant by recoverable and non-recoverable schedules.
6. Explain the role of locks in concurrency control.
7. Describe the use of timestamps in concurrency control.
8. Explain the concept of deadlock and methods to handle it.
9. Describe the shadow-paging recovery method.
10. Explain the concept of deferred and immediate database modification in recovery.
11. **Solve examples on conflict Serializability**

UNIT NO. V

1. Explain the concept of a distributed database system.
2. Describe the CAP theorem with its implications.
3. Discuss the need for NoSQL databases.
4. Compare SQL and NoSQL databases.
5. Explain the BASE properties of NoSQL systems.
6. Describe different types of NoSQL databases with examples.
7. Explain CRUD operations in MongoDB with syntax.
8. Describe how indexing works in MongoDB.
9. Explain the use of aggregation in MongoDB.
10. Discuss the advantages and disadvantages of NoSQL databases

UNIT NO.VI

1. Explain the concept of database connectivity in applications.
2. Describe the role of drivers in database connectivity.
3. Explain the steps involved in connecting a python application to a MySQL database.
4. Describe the function of the DriverManager class.
5. Explain how the Connection interface is used.
6. Describe the use of PreparedStatement interface with example.
7. Explain the purpose of the ResultSet interface.
8. Describe the role of ResultSetMetaData interface.
9. Explain how DatabaseMetaData provides information about a database.
10. Describe the process and use of batch processing in python.

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